

Process Development Report: Selective Laser Melting of Ti6Al4V Femoral Stems

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Executive Summary

This report documents the development and optimization of the Selective Laser Melting (SLM) process for manufacturing Ti6Al4V femoral stems. The process has been successfully scaled to production batches of 20 stems per build, achieving consistent geometric accuracy and mechanical properties while optimizing material utilization and process efficiency. The development focused on parameter optimization, gas flow management, and powder handling protocols to establish a robust manufacturing process suitable for medical device production.

1. Introduction and Project Background

The development initiative was launched to establish an additive manufacturing capability for orthopedic implants, specifically targeting the production of femoral stems from medical-grade Ti6Al4V alloy. The project objectives included developing a repeatable SLM process that meets stringent medical device requirements while maintaining economic viability through optimized material usage and energy efficiency.

The femoral stem geometry presents particular challenges for SLM processing, including complex support structures, varying cross-sectional thicknesses, and the requirement for high-density, defect-free material properties. This report details the progression from initial feasibility studies through to the current production-ready process configuration.

2. Process Development Methodology

2.1 Development Phases

The process development followed a structured approach across three distinct phases:

- Phase 1: Parameter Screening** - Initial parameter optimization using test coupons and simple geometries
- Phase 2: Component Scaling** - Adaptation of parameters to full-scale femoral stem geometry
- Phase 3: Batch Optimization** - Multi-component build optimization and process validation

2.2 Equipment Configuration

The development was conducted using an SLM 280HL system equipped with a 400W ytterbium fiber laser. Key system specifications include: - Build envelope: 280 × 280 × 365 mm - Layer thickness: 30-90 μm - Laser spot size: 80 μm - Inert gas system: Dual-mode argon supply

3. Process Parameter Optimization

3.1 Laser Parameter Development

The laser parameter optimization focused on achieving optimal melt pool characteristics while minimizing spatter and thermal stress. After extensive testing across multiple parameter combinations, the following parameters were established for femoral stem production:

Final Laser Parameters: - Laser power: 400 W - Scan speed: 1200 mm/s - Hatch distance: 120 µm - Layer thickness: 60 µm - Stripes scanning strategy with 67° rotation between layers

The 400W laser power setting provided the optimal balance between melt pool stability and minimal keyhole formation, particularly important for the critical load-bearing regions of the femoral stem.

3.2 Gas Flow Optimization

The inert gas management system underwent significant refinement to address issues with smoke and spatter accumulation observed during initial trials. The final gas consumption values represent an optimized balance between process protection and operational efficiency.

Gas Consumption Summary:

Gas Application	Quantity	Purpose
Chamber Flooding	3.03 kg	Initial oxygen displacement
Building Phase	25.94 kg	Continuous process protection

Note: Historical development data showed initial gas consumption of approximately 35 kg total during early optimization phases. The current configuration represents a 17% reduction through flow rate optimization.

4. Material Flow and Utilization

4.1 Powder Management

The material handling protocol was developed to ensure consistent powder quality and maximize recyclability. The process utilizes gas-atomized Ti6Al4V powder with particle size distribution of 15-45 µm, selected for optimal flow characteristics and melt behavior.

Material Balance for 20-Stem Production Batch:

Material Flow	Quantity	Notes
Fresh powder input	20.83 kg	GA Ti6Al4V, medical grade
Finished stems (including supports)	1.77 kg	Net product weight
Recyclable powder	18.99 kg	Sieved and blended for reuse
Support structures to recycling	0.019 kg	External recycling partner
Filter-captured powder to landfill	0.0208 kg	Fine particulate waste

4.2 Powder Recycling Protocol

The development established a rigorous powder recycling protocol that maintains chemical composition and particle size distribution within specification limits. The 91.2% powder recovery rate represents a significant improvement over the initial development phase recovery of approximately 85%.

The minimal support structure waste (0.019 kg) reflects the extensive optimization of support design to reduce material consumption while maintaining adequate component anchoring and heat dissipation.

5. Energy Consumption Analysis

The energy monitoring throughout development revealed opportunities for optimization, particularly in pre-heat cycles and standby energy management.

Energy Consumption Data:

Process Phase	Energy Consumption	Duration
Total build process	147.26 kWh	61.35 hours
Average power	2.4 kW	Continuous operation

The energy efficiency has improved by approximately 12% compared to early development builds, primarily through optimization of pre-heat parameters and reduction of non-productive machine time.

6. Build Configuration and Productivity

6.1 Nesting Strategy

The development of the 20-stem build configuration required careful consideration of thermal management and geometric constraints. The final nesting pattern ensures adequate spacing between components to prevent thermal interference while maximizing build chamber utilization.

Build Configuration Details: - Components per build: 20 femoral stems - Total build time: 61.35 hours - Build platform utilization: 78% - Z-height utilization: 65%

6.2 Production Rate Analysis

The established process yields a production rate of approximately 0.33 stems per hour of machine time. This represents a 22% improvement over the initial multi-component build strategy, achieved through optimized scan path planning and reduced non-laser time.

7. Quality and Process Control

7.1 In-Process Monitoring

The developed process incorporates multiple in-process monitoring systems: - Layer-wise optical monitoring for powder bed quality - Melt pool monitoring for process stability - Temperature distribution mapping - Oxygen level monitoring (<100 ppm maintained)

7.2 Post-Process Validation

All components undergo comprehensive post-process inspection including: - Dimensional verification (CMM) - Density measurement (Archimedes method) - Surface roughness analysis - Microstructural evaluation - Mechanical testing (representative samples)

The process consistently achieves material density >99.8% and meets all dimensional tolerances specified for femoral stem applications.

8. Challenges and Solutions

8.1 Support Structure Optimization

Early development builds experienced issues with support structure failure and difficult removal. The solution involved: - Implementing variable density support structures - Optimizing support-contact interface geometry - Developing specialized removal tools and procedures

8.2 Thermal Management

The high density of components initially led to thermal accumulation issues, addressed through: - Implementation of staggered build sequence - Optimization of inter-layer delay times - Development of thermal simulation models for build planning

8.3 Powder Handling

Initial powder degradation issues were resolved through: - Implementation of controlled atmosphere powder handling - Development of powder recycling and blending protocols - Installation of dedicated powder sieving and characterization equipment

9. Process Economics

The developed process demonstrates favorable economics for small-batch production of complex orthopedic components. The high material utilization rate (91.2% powder recovery) significantly reduces raw material costs compared to conventional manufacturing approaches.

The 61.35-hour build time for 20 components represents a production efficiency that supports competitive manufacturing costs while maintaining the design flexibility advantages of additive manufacturing.

10. Conclusions and Recommendations

The SLM process for Ti6Al4V femoral stems has been successfully developed and optimized for production batches of 20 components. The process demonstrates robust performance with consistent quality output and optimized resource utilization.

Key Process Metrics Achieved: - Material utilization efficiency: 91.2% powder recovery - Energy consumption: 147.26 kWh per 20-stem batch - Production rate: 20 stems per 61.35-hour build cycle - Quality consistency: >99.8% density, dimensional compliance

Recommended Next Steps: 1. Implement the process for full-scale production 2. Develop automated post-processing procedures 3. Extend the parameter set to additional implant geometries 4. Establish long-term powder recycling performance monitoring

The developed process provides a solid foundation for the additive manufacturing of orthopedic implants, balancing technical requirements with manufacturing efficiency.

Appendices - Appendix A: Parameter Development History - Appendix B: Quality Test Results - Appendix C: Powder Characterization Data

Distribution: Manufacturing Engineering, R&D Department, Quality Assurance, Production Management